DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

Only wake the experts you need

Issue #35 | Tue, 03 Nov 2026 | ☕ ~9 min read

⚡ **If you only have one minute**

1. **Anthropic's updated IPO filing shows about $17B of revenue in one quarter**, a $23.29B loss, and investors pushing the price down to $1.5 trillion.
2. **Meituan (yes, the food-delivery app) launched LongCat-2.5**, a 1.6-trillion-parameter model that only uses about 48B of them for each word.
3. **Part 1: mixture-of-experts**, the trick behind LongCat, Kimi and most giant open models. Big brain, small bill.

Today in three numbers

$16.8B
Anthropic's **Q3 revenue**, up 45% on the quarter before (per WSJ)

48B of 1.6T
parameters **LongCat-2.5 actually uses** for each token

0.016%
of a small model's weights **GPT-6 Astra edited by hand** to fix it

PART 1 Learn one thing

🧑‍🏫 Mixture-of-experts: big models that only use a slice of themselves

4 MIN READ | MODULE 5 · UNDER THE HOOD

A mixture-of-experts (MoE) model is many smaller sub-networks plus a little "router" that picks just two or three of them for each word. You get the knowledge of a giant model at the running cost of a much smaller one.

The plain-English version. In Issue #33 we met the transformer: layers of attention followed by a block of maths called the **feed-forward network**, where much of the model's knowledge lives. A normal, **dense** model runs every token through all of that block. **An MoE model splits it into many "experts"** and adds a **router** that scores them and sends each token to only the top few.

So a model can have **1.6 trillion parameters in total** but only touch about 48 billion for any one token. **Total parameters decide how much it can know; active parameters decide how fast and cheap it runs.** That's why launch posts now quote two numbers.

Analogy: a big Kota coaching institute. It has 64 teachers, but a student's doubt doesn't go to all 64. **The front desk sends it to the two teachers best suited to it**: one for organic chemistry, one for exam tricks. The

institute knows everything; each doubt costs only two teachers' time. The front desk is the router. And if the desk keeps sending everyone to the same star teacher, **the queue explodes**: that's the load-balancing problem.

One token through one MoE layer



Illustrative, using Mixtral's setup: 8 experts, top-2 per token. Real models repeat this in every layer, and the router can pick different experts at each one.

A real example. Mistral's **Mixtral 8x7B** (2023) made MoE famous in open models: 8 experts per layer, 2 chosen per token. Mistral says it has **46.7B parameters in total but uses only 12.9B per token**, so it runs at roughly the speed and cost of a 13B model. (Mistral)

🧱 Dense model		🧩 Mixture-of-experts
• **Every parameter** works on every token	VS	• Router picks **a few experts** per token
• Cost per token grows with model size		• Cost tracks **active** parameters, not total
• Simpler to train and serve		• But **all experts must sit in memory**

Simplified. Experts don't line up neatly with topics like 'maths' or 'Hindi'; they tend to specialise in patterns the router finds useful.

The catch. Active parameters save compute, **not memory**. All 1.6 trillion parameters still have to be loaded, because you don't know which experts the next token needs. That's why one guide to running Kimi K3.1 Pro locally warns you'll need to **clear about 1.6 TB of disk**. (RunAIHome)

Why now? Today's LongCat-2.5 is an MoE, and **Kimi K3.1 Pro's open weights are due tomorrow (4 Nov)**. (Let's Data Science) When you read "1.6T, 48B active", you now know which number matters for your bill.

📖 Jargon buster

- **Expert:** one of many small feed-forward networks inside a layer. Only **a few run per token**.
- **Router (or gate):** a tiny network that **scores the experts** and picks the top-k for each token.
- **Active vs total parameters:** active = **what one token actually uses**; total = everything stored.
- **Load balancing:** an extra training rule that stops the router **overusing a few favourite experts**.

🧪 Try it in 10 minutes

- **No code:** read the first half of Maarten Grootendorst's **visual guide to MoE**. Its pictures of a router choosing experts make this click in minutes. (Exploring Language Models) Then look at any new open model's name: **"A3B" or "48B active"** is the MoE tell.
- **Some code:** paste the toy layer below into a free Colab and run it. **Change `top_k` to 1 or 4** and watch which experts each token picks.

🛠️ For devs: a toy MoE layer in 20 lines

```
import torch, torch.nn as nn, torch.nn.functional as F
class TinyMoE(nn.Module):
    def __init__(self, dim=16, n_experts=8, top_k=2):
        super().__init__()
        self.router = nn.Linear(dim, n_experts)   # scores every expert
        self.experts = nn.ModuleList(nn.Linear(dim, dim) for _ in range(n_experts))
        self.top_k = top_k
    def forward(self, x):                         # x: (tokens, dim)
        weights, picks = self.router(x).topk(self.top_k, dim=-1)
        weights = F.softmax(weights, dim=-1)      # mix the chosen few
        out = torch.zeros_like(x)
        for slot in range(self.top_k):
            for e, expert in enumerate(self.experts):
                mask = picks[:, slot] == e        # tokens sent to expert e
                if mask.any():
                    out[mask] += weights[mask, slot, None] * expert(x[mask])
        return out, picks
out, picks = TinyMoE()(torch.randn(5, 16))
print(picks)   # which 2 of 8 experts each of 5 tokens used
```

Gotchas: for hosting, **size your GPUs by total parameters** (memory) and **your throughput by active ones**. MoE models are also harder to fine-tune: routers can collapse onto a few experts, which is why recipes add a load-balancing loss. Real experts are full feed-forward blocks, not single linear layers.

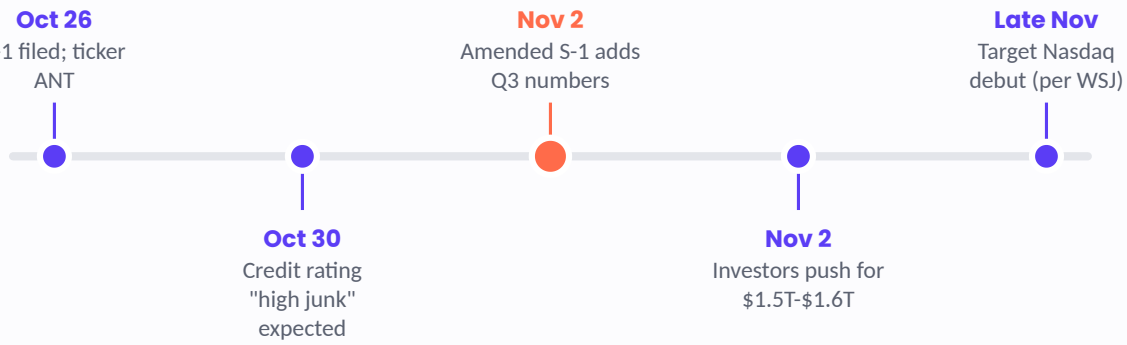
🔗 GO DEEPER

- **Mixture of Experts Explained** (Hugging Face). The standard explainer: history, routing, load balancing, fine-tuning and serving.
- **A Visual Guide to Mixture of Experts** (Maarten Grootendorst). Dozens of diagrams; the best start if you think in pictures.
- **makeMoE: a sparse MoE from scratch** (Hugging Face community blog). Builds a tiny character-level MoE language model in PyTorch, step by step.
- **Mixtral of Experts paper** (Mistral AI, arXiv). Short and readable; see how routing behaves on real text.

PART 2 What moved in AI

7 min read · Covering the last 24 hours. Anthropic's real numbers, a delivery giant's trillion-parameter model, and an AI that fixed another AI by editing its brain directly.

Anthropic's road to the stock market



Top story 1 · Update: Anthropic's IPO filing shows a $17B quarter and a $23B loss

BIG DEAL MONEY

What happened. Anthropic filed an **amended S-1**, the document a company must publish before listing shares, and this time it includes **July to September numbers** (Issue #28 covered the first filing). CNBC reports revenue of **roughly $17 billion for the quarter** and a **net loss of $23.29 billion.** (AI Weekly, via CNBC)

Anthropic revenue in 2026: one quarter beats a half-year ($ billion)



Six-month figure from the first S-1 (Issue #28). Q3 is 'roughly $17B' per CNBC and $16.8B per WSJ, both via AI Weekly; nine-month total $33.24B.

The details that matter:

- **The loss is mostly paperwork:** per CNBC's summary, it's driven mainly by **changes in the value of convertible notes and derivatives**, not by running costs alone.
- **The compute bill:** Anthropic lists about **$895.4 billion in purchase commitments and leases**, "substantially all" for compute and infrastructure.
- **The price cut:** big investors told Anthropic to list at **$1.5 to $1.6 trillion**, down from roughly $2 trillion floated earlier, after last week's tech selloff and to leave room for a first-day pop. (AI Weekly, via WSJ)
- **Timing:** the roadshow could start **as soon as next week**, with a Nasdaq debut targeted for **the week before Thanksgiving** (26 Nov).
- **A strange stat:** Business Insider reports that Anthropic staff's own Claude use, **priced at retail API rates**, now exceeds what it pays employees. Commenters note it isn't real cash spent. (AI Weekly, via Business Insider)

Why it matters. These are the most detailed numbers yet on **how fast money for AI coding and agents is growing**: about 45% in a single quarter. The bets are bigger still: **nearly $900B in commitments** against a ~$17B quarter.

WHO SHOULD CARE 🙋 Everyone ●●●○○ 🎨 Vibe coders ●●○○○ 🛠️ Developers ●●○○○
🔥 Hype level ●●●●○

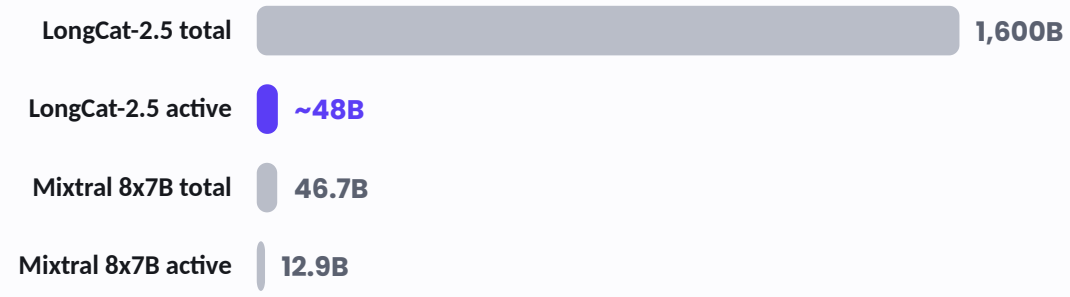
👀 **Should you care?** If you build on Claude, a well-funded Anthropic is good news for **uptime and prices**. If you're thinking of buying the stock via a US brokerage app, remember that **IPO prices and first-day pops are not the business**. (This isn't financial advice.)

Top story 2 · Meituan's LongCat-2.5: a 1.6-trillion-parameter agent model

NEW MODEL FOR DEVS

What happened. Meituan, China's food-delivery giant, released **LongCat-2.5 Preview**. It's a natively multimodal **mixture-of-experts** model with **1.6 trillion parameters in total, about 48 billion active per token**, and a **1-million-token context window**. It's live in Meituan's chat app and API now. (AI Weekly, via TechNode)

Total vs active parameters (billion)



LongCat per TechNode via AI Weekly; Mixtral per Mistral (2023). LongCat uses about 3% of its weights per token (our maths).

The details that matter:

- **Built for long jobs:** it's pitched at **agents that work for hours** across terminals, browsers, spreadsheets and design tools. Meituan says it trained on **long task sequences** and turned **real agent failures into practice tasks**.
- **Sees and reads together:** images and text go through **one shared token stream**, aimed at visual coding, design and 3D work.
- **Homegrown chips:** Meituan says it was trained on **domestic Chinese AI accelerators**. It hasn't said which ones.
- **Weights later:** GitHub and Hugging Face releases are **planned, with no date or licence yet**. Meituan's benchmark claims haven't been independently checked.

Why it matters. A **food-delivery company** now ships trillion-parameter models. MoE is what makes that affordable to run, and China's open-model race keeps adding players, **many of them releasing weights**.

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🔥 Hype level ●●●○○

👀 **Should you care?** Not yet, unless you test models for agent work. **Wait for the weights and an independent benchmark.** Kimi K3.1 Pro's weights (due tomorrow) are the nearer one to watch.

Top story 3 · GPT-6 Astra fixed a small model by editing its weights directly

RESEARCH FOR DEVS

What happened. MIT CSAIL researchers (Rikiya Takehi, Jayden Teoh, Alex Zhang and Omar Khattab) describe **"weight-space programming"**: **GPT-6 Astra reads and edits another model's weights** from a plain-English spec, with **no gradient descent** (the usual way training nudges weights). (AI Weekly)

Qwen3.5-0.8B after the edit (score %, before → after)



Authors' own results, as summarised by AI Weekly. Not yet independently reproduced.

The details that matter:

- **The bug:** Alibaba's tiny **Qwen3.5-0.8B** kept getting stuck **repeating itself**. Astra traced the looping to specific **directions in layers 15 and 19**.
- **The fix:** it removed those directions, changing **just 0.016% of the parameters**, and scores jumped on maths and instruction-following tests.
- **The source:** the write-up is on the lead author's **project blog**. We couldn't open it ourselves, so these details come via AI Weekly's summary.

Why it matters. It's like Issue #34's training story run backwards: instead of **training a behaviour in**, a model **found a bad behaviour and cut it out**. If it holds up, it links interpretability (looking inside models) to **practical repair**.

WHO SHOULD CARE 🙋 Everyone ●○○○○ 🎨 Vibe coders ●○○○○ 🛠️ Developers ●●●○○
🔥 Hype level ●●●○○

👀 **Should you care?** A research curiosity for now. If a small local model **loops**, try `repetition_penalty` before swapping models.

⚡ Quick hits

📊 GDPx: best model scores 12%.

SemiAnalysis's new agent benchmark across 11 fields puts **Claude Opus 5.5 at 12.0%**, Gemini 4 Carbon 10.9%, GPT-6 Astra 9.6%. Finance and legal: **0% for everyone**. (AI Weekly)

🏦 Goldman pulls coding agents from bankers.

Per Bloomberg, **Claude Code and GitHub Copilot** are now limited to Goldman's **~12,000 engineers**, after an October review. (AI Weekly)

🌐 Squarespace cuts 19%.

About **300 jobs**, per TechCrunch, as it readies **Forma**, an "AI-native" site builder. The company didn't blame AI. (AI Weekly)

🔒 One paid ChatGPT per device.

OpenAI now blocks stacking paid plans on **one device or card** ("Device limit reached"), per The Decoder. (AI Weekly)

🧪 QA agents, tested.

Momentic's **Mo** found **269 of 1,396** labelled bugs in an hour on its own benchmark, vs 86 for Codex and 59 for Claude Code. Costlier per bug than Codex. (AI Weekly)

📋 Update: shared ChatGPT/Codex limits.

OpenAI's help page now explains the **December shared allowance**; ordinary Instant chats don't count against it (Issue #34). (The Neuron)

🔌 Marvell buys Lumentum.

A **$115B all-stock deal** for the optical parts that link AI data-centre chips, per CNBC. (AI Weekly)

🗓️ Coming up.

Kimi K3.1 Pro weights and Apple's China beta **4 Nov**. Ollie shuts **6 Nov**. US lame-duck session **9 Nov**. Anthropic's new usage policy **12 Nov**. Anthropic's IPO, **before Thanksgiving** (per WSJ).

🎯 One thing to try today: read a model card like a pro

Open any new open model on Hugging Face and find its `config.json`. Look for `num_experts` (Mixtral calls it `num_local_experts`) and `num_experts_per_tok`. Total experts tell you **the memory you'll need**; experts per token tell you **the speed you'll get**.

🧠 Pop quiz: 30 seconds, no Googling

1. In an MoE model, what decides which experts a token uses?

- A The user
- B A router
- C Random chance

2. LongCat-2.5 has 1.6T parameters. About how many run per token?

- A 1.6T
- B 480B
- C 48B

3. What mostly drove Anthropic's $23.29B Q3 loss?

- A Salaries
- B Note and derivative revaluations
- C Refunds

Answers (no peeking): 1. B: it scores experts and picks the top few · 2. C: about 3% of the total · 3. B: per CNBC's summary of the filing

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